



ABSTRACT BOOK

NAACCR 2026 Virtual Conference

True North: Aligning Data, Discovery, and Impact

June 23 - 25, 2026





Concurrent 3.A National Childhood Cancer Registry (NCCR) Research Use Cases

June 24, 2026, 2:30 PM - 3:45 PM

Using the NCCR Data Platform to Identify Predisposition Syndromes in Pediatric Cancer Patients

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Background

Genetic predisposition syndromes and congenital conditions, such as Fanconi anemia, Down syndrome, and Bloom syndrome, are known to increase the risk of specific pediatric cancers. However, information on these conditions is often incomplete or unavailable in population-based cancer registry data because data items that capture secondary diagnoses are restricted, limiting the ability to study their prevalence and distribution at the population level.

Purpose

The National Childhood Cancer Registry (NCCR) Data Platform provides a unique opportunity to study these associations by linking population-based cancer registry data with medical claims diagnosis data. The claims source includes International Classification of Diseases (ICD-9 and ICD-10) codes, sequence numbers for diagnoses occurring on multiple claims, and timing relative to the index cancer diagnosis. These ICD diagnosis codes will be used directly to identify documented genetic and congenital predisposition syndromes that precede or co-occur with pediatric cancer diagnoses. Using the NCCR Data Platform, this study aims to describe distribution of genetic predisposition syndromes identified through linkage between pediatric cancer cases and secondary diagnosis codes recorded in medical claims data.

Methods/Results

The NCCR Data Platform includes linkage of Surveillance, Epidemiology, and End Results (SEER) population-based cancer registry data from 1995 to 2022, representing 52.7% of all U.S. children and adolescents and young adults (AYA), with a medical claims diagnosis data source containing ICD-coded diagnoses from multiple insurers over the same period. Genetic predisposition syndromes will be identified using specific ICD-9 and ICD-10 codes recorded in medical claims. Descriptive analyses will summarize the frequency and distribution of identified genetic syndromes across pediatric cancer types and demographic characteristics.

Conclusion

Findings will provide new insights into cancer etiology and contribute to a better understanding of the relationship between pediatric cancers and underlying predisposition conditions. This work highlights the value of the NCCR Data Platform in generating original and clinically meaningful information to support cancer surveillance and research.